PRITAM KUMAR PANDA, PhD.

AI Drug Discovery Research Scientist | 2026

CONTACT INFORMATION

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LinkedIn | Portfolio | GitHub | Google Scholar

RESEARCH FOCUS

AI-Driven Drug Discovery; Protein Foundation Models; Protein Design; Molecular Docking & Virtual Screening; Molecular Dynamics & Free-Energy Calculations; Physics-Based Molecular Modeling; Active Learning; Cheminformatics; Scientific Software & Algorithm Development

EXECUTIVE PROFILE

Drug Discovery Research Scientist with 8+ years of experience spanning molecular modeling, protein foundation models, molecular docking, molecular dynamics, free-energy methods, and scientific software development. Builds physics-informed and machine-learning-driven platforms that accelerate structure-based drug discovery and therapeutic design. Experienced in developing new docking algorithms, training and deploying protein models, implementing FEP and MD workflows, and scaling computational research across GPU/HPC environments.

WORK EXPERIENCE

Stanford University 2025–Present

Postdoctoral Scholar – AI-Driven Drug Design & Molecular Modeling *Stanford, CA*

I develop computational methods for AI-driven drug discovery, combining protein foundation models, molecular docking, molecular dynamics, and physics-based free-energy calculations. My work focuses on integrating machine-learning models with molecular simulation to prioritize and optimize therapeutic candidates.

- Advanced a DoD/DARPA anesthesia program to Phase II clinical trials by architecting an AI-driven drug discovery platform using machine learning techniques that screened 2.5B small molecules and generated 100,000+ AlphaFold-based receptor conformations.
- Cut experimental candidate testing 60% by implementing GPU-accelerated atomistic simulation workflows coupled with active learning-based docking triage and ADMET/QSAR profiling.

Stack: *Python; PyTorch/JAX; RDKit; AlphaFold2/3; Schrödinger; GROMACS; OpenMM; AutoDock-GPU; Vina; Glide; CDOCKER; Boltz-2; AWS/HPC; CUDA*

German Cancer Research Center (DKFZ) 2023–2024

Bioinformatician – HPC Pipeline Development & Cloud Infrastructure *Heidelberg, Germany*

I orchestrated and optimized reproducible scientific computing workflows across HPC and cloud environments, combining Nextflow, Docker/Singularity, CI/CD, GPU acceleration, and large-scale job orchestration to support high-throughput biomedical research. My work focused on improving computational efficiency, workflow reliability, and reproducibility while building research infrastructure that could scale across multidisciplinary teams and large user communities.

- Reduced analysis runtime by ~40%, lowered pre-production software defects by ~95%, and supported 200+ concurrent computational jobs through workflow optimization, validation, and automation.
- Developed performance-profiled GPU workloads and contributed to the HI-TRON FAIR data portal, reducing compute costs by ~35% and supporting reproducible research infrastructure used by 1,000+ scientists.

Stack: *Nextflow; Snakemake; Bash; Docker/Singularity; HPC/SLURM; Vue.js; R; Python; GitLab/GitHub; CI/CD*

Karolinska Institute 2021–2023

Affiliated Researcher – Biomarker Discovery & Clinical ML *Stockholm, Sweden*

I developed predictive and immunoinformatics workflows integrating antibody-profiling, molecular, genomic, and clinical data to support patient stratification, antibody modeling, and therapeutic hypothesis generation. My work focused on translating heterogeneous clinical and molecular datasets into computational models that could inform biomarker discovery and therapeutic research.

- Developed clinical machine-learning models for patient stratification using genomic and antibody-profile data, contributing to 2 peer-reviewed publications and 1 patent.
- Engineered immunoinformatics workflows for antibody modeling and analysis across 2TB+ multimodal clinical datasets, supporting research activities associated with 3 ongoing clinical trials and collaboration with Inflanova AB, Sweden.

Stack: *Python; R; scikit-learn; immunoinformatics pipelines; clinical genomics/antibody-profiling data*

Uppsala University 2018-2023
Doctoral Researcher – Computational Biophysics & Quantum Physics *Uppsala, Sweden*

My Ph.D. research focused on developing multiscale computational frameworks spanning molecular dynamics, enhanced sampling, QM/MM, quantum chemistry, and electronic-structure modeling to study proteins, membrane systems, drug–target interactions, nanopore sensing, and bio-inspired materials. My work combined physics-based simulation with scientific software development and HPC automation, resulting in 5 peer-reviewed publications, international collaborations, and reusable computational workflows across multiple research projects.

- Originated GENOME2QUNOME, a quantum-to-biological modeling framework connecting DFT-based 2D-material simulations with atomistic biological modeling, and led ab initio electronic and thermal-transport studies that informed material-selection research with Hitachi ABB.
- Implemented replica-exchange and umbrella-sampling protocols together with Python-based automation for 128-node HPC clusters, reducing simulation time by ~55% and standardizing molecular-simulation workflows across collaborative research projects.

Stack: *VASP; SIESTA/TranSIESTA; DFT; Python; ML-augmented electronic-structure modeling*

Freiburg Medical Center 2017-2018
NGS Analyst – Pediatric Hematology & Oncology *Freiburg, Germany*

I developed and validated high-throughput WES/WGS and targeted cancer-panel pipelines for clinical hematology and oncology, with a focus on accurate variant detection, annotation, and reproducible analysis for myelodysplastic syndromes and related hematologic disorders.

- Engineered and optimized WES/WGS variant-calling and RNA-seq workflows processing 200+ clinical samples/month, reducing manual runtime by ~70% and improving alignment performance by ~2.5x through HPC parallelization and workflow optimization.
- Contributed computational analyses to the European MDS Working Group, supporting a Nature Medicine publication on clonal hematopoiesis trajectories and disease evolution.

Stack: *BWA-MEM; STAR; GATK; ANNOVAR/VEP/CADD; HPC/SLURM; RNA-Seq*

EDUCATION

Ph.D., Physics 2018–2023
Uppsala University *Sweden*
Dissertation Focus: Physics-based molecular modeling, molecular dynamics, free-energy calculations, and multiscale simulations of proteins and biomolecular systems

MTech., Bioinformatics 2014–2016
D Y Patil University *India*

M.Sc., Bioinformatics 2012–2014
Utkal University *India*

HONORS & RECOGNITION

- *Best Poster Award,* Stanford University School of Medicine, 2026
- Fellow of *Thinking About Thinking*, 2026
- Elected Member of *Sigma Xi,* The Scientific Research Honor Society, 2025
- *Nextflow Ambassador*, 2024-present
- Invited as a guest speaker by leading *TV news channels* in India, 2023
- Featured in *The Global Indian Magazine* "Indians in Europe", 2023
- Recognized by Elsevier and Springer as an *outstanding reviewer*, 2022-2023
- Awarded *research grant* from Colgate-Palmolive for CADD project, 2022
- Two-time *Chairperson* at the e-Materials Research Society conference, 2022
- Awarded IMM Strategic Interdisciplinary *Collaboration grant* at Karolinska Institute, 2022
- Awarded Uppsala University *innovation grant* by ABB Power Grids, Hitachi, Sweden, 2021

SELECTED PUBLICATIONS

- Structure-Based Drug Designing and Immunoinformatics Approach for SARS-CoV-2.
  Pritam Kumar Panda, *Murugan Natarajan Arul, Paritosh Patel, Suresh K. Verma, Wei Luo, Horst-Günter Rubahn, Yogendra Kumar Mishra, Mrutyunjay Suar, Rajeev Ahuja.*
  Science Advances*, 2020*

- Alchemical Free Energy Perturbation Predicts Relative Binding Affinities of Propofol Analogs and Etomidate Stereoisomers at the $GABA_AR$. ***Pritam Kumar Panda***, *Edward J. Bertaccini*. ***ACS Omega****, in revision, 2026*
- Structural Context Determines Docking Engine Performance: A Family-Stratified Benchmark of Six Engines. *Kristoffer Alejo, Sarah Fisher, Tejaswan Kalluri, Balkrushna More, Hrishikesh Rajgure,* ***Pritam Kumar Panda,*** *Christopher Korban, Christian Chung. Preprint, 2026 · DOI: 10.64898/2026.08.11.744016*
- PandaMap: A Python Package for Comprehensive Visualization of Protein-Ligand Interaction Networks. ***Pritam Kumar Panda***. bioRxiv, *2026 · DOI: 10.64898/2026.08.06.743421*
- PandaFEP: Free Energy Perturbation Engine for Alchemical Binding Free-Energy Calculations, Built from First Principles. ***Pritam Kumar Panda***. *Manuscript in review, 2026*
- PandaDock: An Open-Source Molecular Docking Platform with Flexible-Ligand Search and Equivariant Neural Scoring. ***Pritam Kumar Panda***. *Manuscript in review, 2026*

PATENTS & IP

Composition, Methods and Uses
Johan Frostegård, Shailesh Kumar Samal, ***Pritam Kumar Panda***
U.S. Patent Application 18/864,303*, 2025*

ADVISORY

Scientific Advisor — 2025–Present
Revilico Inc. — *Los Angeles, CA*
Computational drug discovery / scientific roadmap and evaluation frameworks

Bioinformatics Research Consultant — 2021–2023
Colgate-Palmolive — *Remote, United States*
Computational drug discovery / dermatology / publication impact.

Founder & CEO — 2021–2023
Nerdalytics — *Uppsala, Sweden*
Research consulting, clients, scientific strategy, delivery, and business impact.

TECHNICAL LEADERSHIP

AI/ML: PyTorch, TensorFlow, JAX, scikit-learn, NVIDIA NeMo, protein foundation models, geometric deep learning, equivariant neural networks, statistical modeling

Drug Discovery: AlphaFold2/3, OpenFold, Boltz-2, RFdiffusion, GROMACS, OpenMM, AutoDock-GPU, Vina, Glide, CDOCKER, RDKit, FEP, QM/MM, molecular docking, virtual screening, free-energy calculations

Translational Bioinformatics: WES/WGS, RNA-seq, scRNA-seq, biomarker analysis

HPC/Cloud: NVIDIA GPUs, CUDA/RAPIDS, SLURM, PBS, AWS, Nextflow, Snakemake, Docker, Singularity, CI/CD

Software: Python, R, Bash, FastAPI, Flask, Next.js, React, Plotly, Streamlit, Git

OPEN-SOURCE PROJECTS

PandaDock - GPU-Accelerated Molecular Docking & Virtual Screening
Designed and implemented docking/scoring algorithms for high-throughput virtual screening with CUDA-accelerated execution; ~10× speedup over CPU implementations and open-source workflows for reproducible structure-based discovery.

PandaMap / PandaProt - Protein–Ligand & Interface Analysis
Built residue-level interaction-mapping tools for protein–ligand, protein–protein, protein–nucleic-acid and antigen–antibody systems, supporting binding-site interpretation and structure-guided design.

R-Group Designer / Optimus Chem - Medicinal Chemistry & Lead Optimization
Created RDKit-based tools for analog enumeration, ADMET/drug-likeness analysis, molecular-property prediction, 3D visualization and iterative lead-design workflows.

PROFESSIONAL DEVELOPMENT

NVIDIA – Accelerated Data Science & Foundation Models in Digital Biology | Stanford Online – Machine Learning Specialization | AWS – Technical Essentials / Cloud Practitioner
QIAGEN – Ingenuity Pathway Analysis | EMBL-EBI/GHGA – Nextflow / nf-core

INTERESTS

Calisthenics; Meditation; Gardening; Teaching; Scientific content creation